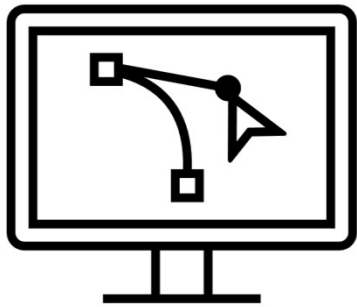


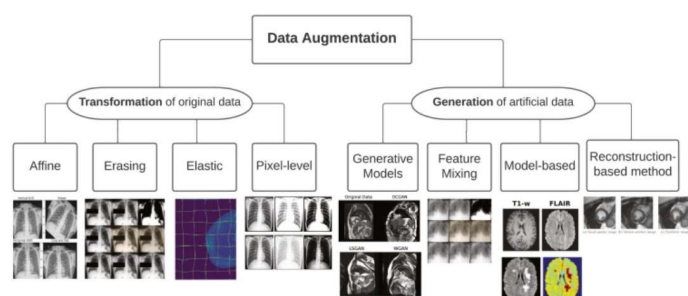


# Data Augmentation

151

## Data Augmentation

- Definition
- Operations: Image, Text & Audio
- Images Commonly Operations:
  - Rotation
  - Flip
  - Zoom
  - Crop
  - Noise Injection
  - Brightness and Contrast Adjustment

Source: ScienceDirect<sup>6</sup>

152

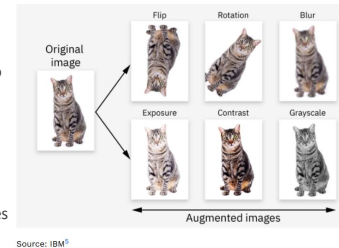
# What is Data Augmentation?

**Data augmentation** is a technique of artificially increasing the training set by creating modified copies of a dataset using existing data. It includes making minor changes to the dataset or using deep learning to generate new data points.

## Augmented vs. Synthetic data

- **Augmented data** is driven from original data with some minor changes. In the case of image augmentation, make geometric and color space transformations like, flipping, resizing, cropping, brightness, or contrast, to increase the size and diversity of the training set.
- **Synthetic data** is generated artificially without using the original dataset. It often uses DNNs (Deep Neural Networks) and GANs (Generative Adversarial Networks) to generate synthetic data.

**Augmentation techniques** are not limited to images. It is possible to augment audio, video, text, and other types of data.

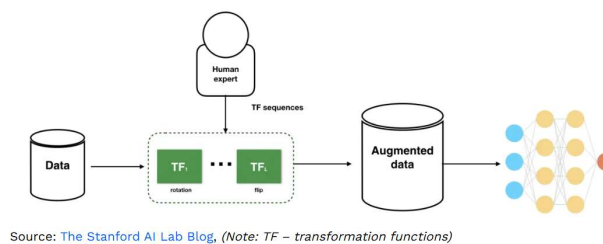


153

153

# When Use Data Augmentation?

- To prevent models from overfitting
- The initial training set is too small
- To improve the model accuracy
- To Reduce the operational cost of labeling and cleaning the raw dataset



154

154

## Data Augmentation limitations

- The biases in the original dataset persist in the augmented data.
- Quality assurance for data augmentation is expensive.
- Research and development are required to build a system with advanced applications. For example, generating high-resolution images using GANs can be challenging.
- Finding an effective data augmentation approach can be challenging.

155

155

## Data Augmentation Techniques

### Audio Data Augmentation

- Noise injection: add gaussian or random noise to the audio dataset to improve the model performance.;
- Shifting: shift audio left (fast forward) or right with random seconds.
- Changing the speed: stretches times series by a fixed rate.
- Changing the pitch: randomly change the pitch of the audio.

### Text Data Augmentation

- Word or sentence shuffling: randomly changing the position of a word or sentence.
- Word replacement: replace words with synonyms.
- Syntax-tree manipulation: paraphrase the sentence using the same word.
- Random word insertion: inserts words at random.
- Random word deletion: deletes words at random.

### Image Augmentation

- Geometric transformations: randomly flip, crop, rotate, stretch, and zoom images. You need to be careful about applying multiple transformations on the same images, as this can reduce model performance.
- Color space transformations: randomly change RGB color channels, contrast, and brightness.
- Kernel filters: randomly change the sharpness or blurring of the image.
- Random erasing: delete some part of the initial image.
- Mixing images: blending and mixing multiple images.

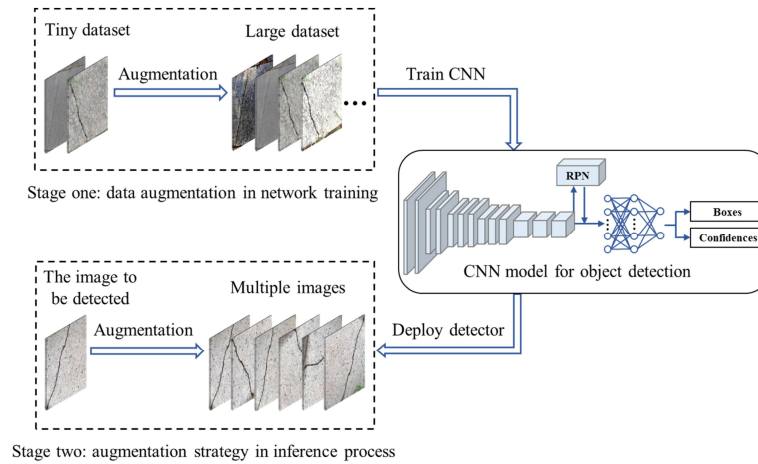
### Advanced Techniques

- Generative adversarial networks (GANs): used to generate new data points or images. It does not require existing data to generate synthetic data.
- Neural Style Transfer: a series of convolutional layers trained to deconstruct images and separate context and style.

156

156

## CNN's: Architecture for Data augmentation



157

## Data Augmentation

It's common data augmentation techniques used in computer vision tasks (such as image classification, object detection, and segmentation) and natural language processing tasks:

-  Is a technique used to artificially increase the size and diversity of a dataset by applying various transformations to the existing data samples.
-  Is commonly used in machine learning, especially in scenarios where the original dataset is limited in size or lacks diversity.
-  Helps to improve the robustness, generalization, and performance of machine learning models by exposing them to a wider range of variations and scenarios.
-  Is typically applied during the training phase of machine learning models, where the augmented data samples are used to train the model.
-  Additionally, can help reduce overfitting, especially when dealing with small or imbalanced datasets.

158

# Operations

## Data Augmentation

### Text Augmentation

- **Synonym Replacement:** Replace words in text with their synonyms to introduce variations in vocabulary.
- **Random Insertion:** Insert random words into sentences to simulate noisy input data.
- **Random Deletion:** Delete random words from sentences to encourage the model to learn to deal with missing information.
- **Random Swap:** Swap the positions of two words in sentences to introduce variations in word order.

### Audio Augmentation

- **Speed Perturbation:** Increase or decrease the playback speed of audio samples to simulate variations in speaking rates.
- **Pitch Shift:** Change the pitch of audio samples to simulate variations in speaker characteristics.
- **Noise Addition:** Add background noise or environmental sounds to audio samples to simulate different recording conditions.

159

# Operations

## Data Augmentation

### Image Augmentation



#### Rotation

Randomly rotate images by a certain angle to introduce variations in object orientations.



#### Flip

Horizontally or vertically flip images to account for variations in object poses.



#### Zoom

Randomly zoom in or out on images to simulate different scales and perspectives.



#### Crop

Randomly crop regions from images to focus on specific objects or regions of interest.



#### Brightness and Contrast Adjustment

Adjust the brightness, contrast, and saturation of images to simulate changes in lighting conditions.



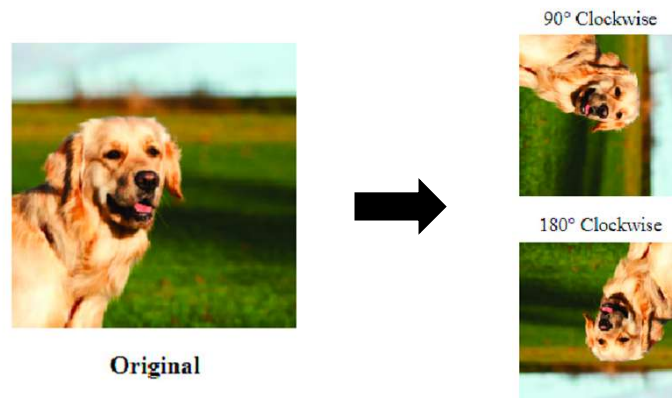
#### Noise Injection

Add random noise (e.g., Gaussian noise) to images to simulate sensor noise or image distortions.

160

## Operation - **Rotation**

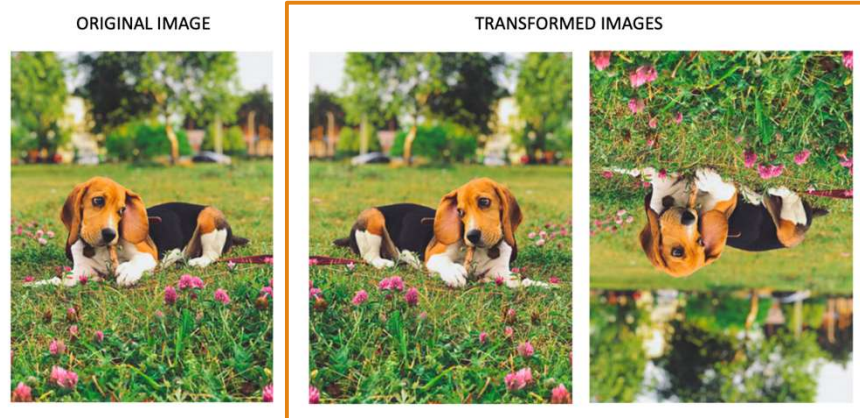
Data Augmentation



161

## Operation - **Flip**

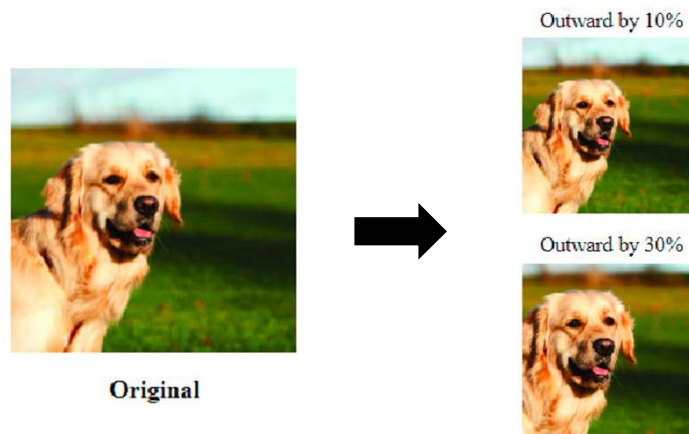
Data Augmentation



162

## Operation - **Zoom**

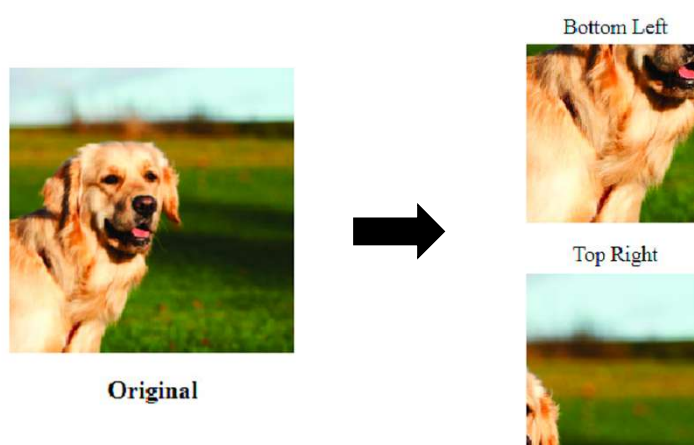
Data Augmentation



163

## Operation - **Crop**

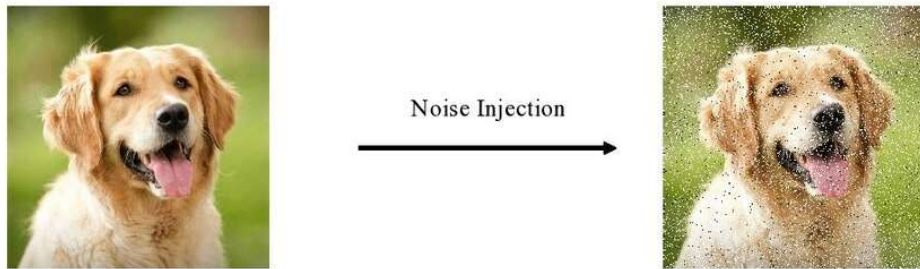
Data Augmentation



164

## Operation - **Noise Injection**

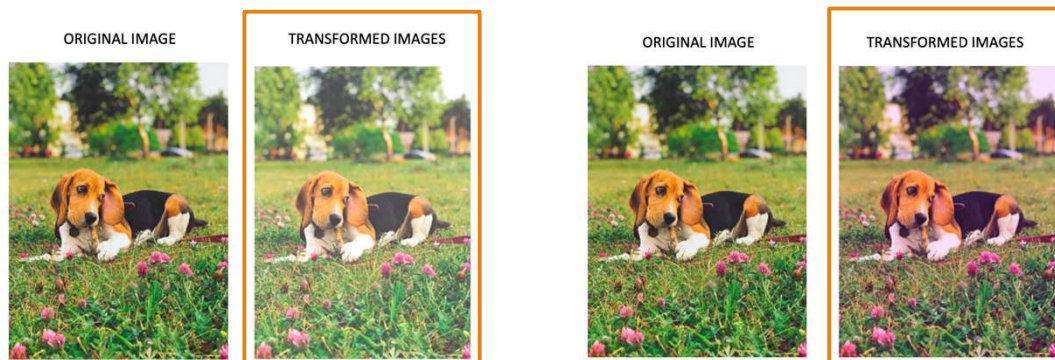
Data Augmentation



165

## Operation - **Brightness and Contrast Adjustment**

Data Augmentation



166

## Advantages

### Data Augmentation



#### Increased Dataset Size

Artificially increases the size of the dataset by generating new samples through transformations.



#### Improved Generalization

By exposing the model to a wider range of variations and scenarios, data augmentation helps improve the model's ability to generalize well to unseen data.



#### Reduced Overfitting

Introduces variations in the training data, which can help prevent the model from memorizing the training examples and overfitting to the training data.



#### Robustness to Input Variability

Models trained with augmented data are more robust to variations and distortions in the input data, such as changes in lighting conditions, object poses, or background clutter.

167

## Disadvantages:

### Data Augmentation



#### Computational Overhead

Data augmentation increases the computational cost of training, as it requires generating and processing additional data samples during training.



#### Risk of Overfitting

While data augmentation can help prevent overfitting in many cases, it's possible to over-augment the data, leading to a model that performs poorly on unseen data. Careful selection and tuning of augmentation techniques are necessary to avoid this issue.



#### Increased Training Complexity

Managing and tuning a diverse set of augmentation techniques adds complexity to the training pipeline.



#### Task-Specific Considerations

Not all augmentation techniques are suitable for every task or dataset. Some techniques may be more effective than others depending on the characteristics of the data and the requirements of the task.

168